

HomeGuard Kerala

A Geo-Verified Property Inspection and Evidence Reporting System For NRI-Owned Home

Abstract:

A secure, geo-verified property inspection and evidence-reporting platform designed to provide reliable oversight for NRI-owned homes. HomeGuard integrates a health-check scheduler for planned inspections, a geo-fenced inspector app that ensures physical presence at the site, and an evidence-based reporting engine that compiles photos, videos, and structured notes into a professional dashboard. The platform standardizes inspection activities, highlights potential risks using red-flag logic, and empowers owners to make timely maintenance decisions. By combining geolocation enforcement, guided workflow, and transparent evidence reporting, HomeGuard Kerala transforms informal property checking into a dependable and professional service while creating part-time work opportunities for local students.

- **Admin:** Manages the entire platform by verifying agents, overseeing inspections, and ensuring that all activities are carried out smoothly and accurately
- **Inspector(Agent):** They receive job tickets, perform geo-verified inspection, must capture clear (photos, videos, and notes of each area) for evidence
- **User(NRI HomeOwner):** Registers property details, selects inspection package (Basic/Deep Check), schedules recurring inspections via the Health-Check Scheduler, and view reports

The system consists of **seven core modules** that together form a complete working MVP

1. OwnerConnect Portal

Module for NRI Homeowners

This module allows NRI homeowners to interact with the system. They can register, add their property details, schedule inspections, and view reports. It acts as the central interface for all user actions.

2. InspectSync Scheduler

Inspection Planning & Job Ticket System

This module manages all inspection schedules. It creates job tickets automatically when a homeowner books a check and assigns inspectors. It tracks inspection status from pending to completion.

3. FieldForce Inspector App

Inspector Workflow & Checklist Module

This module is used by certified inspectors on their mobile devices. It provides them with job tickets, enforces the step-by-step checklist, and allows them to upload photos, videos, and notes during an inspection.

4. GeoShield Verification Engine

GPS-Based Location Validation

This module ensures that inspectors can start an inspection only when they are physically present at the property. It uses geofencing, GPS validation, and geotagged media uploads to prevent fake or remote inspections.

5. ProofVault Evidence Manager

Media Storage & Validation Module

This module securely handles all uploaded evidence such as photos, videos, and comments. It ensures every checklist item is completed and stores the data safely in the database for later report generation.

6. Insight360 Reporting Hub

Automated Report Generation

This module builds a visual and comprehensive inspection report using the collected evidence. It highlights red-flag issues like leaks, cracks, electrical faults, and pest activity, giving the homeowner a clear view of property condition.

7. CommandDesk Admin Console

Administration & Monitoring Module

This module is used by the system administrator to manage inspectors, approve accounts, monitor ongoing inspections, and oversee the overall system performance. It maintains operational control of the platform.

Technology Stack

Frontend: React (Vite) + Tailwind CSS

Backend: FastAPI (Python) for scheduling, rule enforcement, and media processing

Database: PostgreSQL for property records, inspection logs, and red-flag events

Integration: Whatsapp for reminder and report notification